

Differential Effects of the G118R, H51Y, and E138K Resistance Substitutions in Different Subtypes of HIV Integrase

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ABSTRACT

Dolutegravir (DTG) is the latest antiretroviral (ARV) approved for the treatment of human immunodeficiency virus (HIV) infection. The G118R substitution, previously identified with MK-2048 and raltegravir, may represent the initial substitution in a dolutegravir resistance pathway. We have found that subtype C integrase proteins have a low enzymatic cost associated with the G118R substitution, mostly at the strand transfer step of integration, compared to either subtype B or recombinant CRF02_AG proteins. Subtype B and circulating recombinant form AG (CRF02_AG) clonal viruses encoding G118R-bearing integrases were severely restricted in their viral replication capacity, and G118R/E138K-bearing viruses had various levels of resistance to dolutegravir, raltegravir, and elvitegravir. In cell-free experiments, the impacts of the H51Y and E138K substitutions on resistance and enzyme efficiency, when present with G118R, were highly dependent on viral subtype. Sequence alignment and homology modeling showed that the subtype-specific effects of these mutations were likely due to differential amino acid residue networks in the different integrase proteins, caused by polymorphic residues, which significantly affect native protein activity, structure, or function and are important for drug-mediated inhibition of enzyme activity. This preemptive study will aid in the interpretation of resistance patterns in dolutegravir-treated patients.

IMPORTANCE

Recognized drug resistance mutations have never been reported for naive patients treated with dolutegravir. Additionally, in integrase inhibitor-experienced patients, only R263K and other previously known integrase resistance substitutions have been reported. Here we suggest that alternate resistance pathways may develop in non-B HIV-1 subtypes and explain how “minor” polymorphisms and substitutions in HIV integrase that are associated with these subtypes can influence resistance against dolutegravir. This work also highlights the importance of phenotyping versus genotyping when a strong inhibitor such as dolutegravir is being used. By characterizing the G118R substitution, this work also preemptively defines parameters for a potentially important pathway in some non-B HIV subtype viruses treated with dolutegravir and will aid in the inhibition of such a virus, if detected. The general inability of strand transfer-related substitutions to diminish 3' processing indicates the importance of the 3' processing step and highlights a therapeutic angle that needs to be better exploited.

During the past 30 years, human immunodeficiency virus (HIV) has infected >68 million people, killing ~34 million (1). The advent of combination antiretroviral (ARV) regimens in 1995 has played a key role in reducing HIV/AIDS-related deaths and prolonging the life span of people living with HIV (2).

The integration of viral DNA into the human genome prevents the facile eradication of HIV from the cells of infected individuals, even during suppressive ARV therapy (3). Additionally, HIV quickly develops drug resistance in the face of suboptimum drug pressure (as happens in poorly adherent patients or due to low drug bioavailability) (4), and this is facilitated by the high mutation and recombination rates of HIV reverse transcriptase (5). The inability to eradicate HIV combined with high rates of drug resistance have necessitated the ongoing development of new potent ARVs with novel targets, unique resistance patterns, and/or better bioavailability in a context of simplified dosing (6, 7).

Raltegravir (RAL) was the first approved drug to target the HIV integrase (IN) enzyme. RAL acts as an integrase strand transfer inhibitor (INSTI) and was approved in 2007 (8), followed by a second INSTI, termed elvitegravir (EVG), in 2012 (9). These two compounds as well as dolutegravir, the most recently approved

INSTI (2013) (10), target the strand transfer step of integration by three binding mechanisms: (i) chelation of active-site divalent cations (Mg^{2+} or Mn^{2+}), (ii) pi-stacking interactions between INSTI halobenzyl groups and the viral long terminal repeat (LTR) base immediately upstream of CA-OH, and (iii) interactions between INSTIs and specific IN residues (11). Both RAL and EVG are associated with a low barrier to resistance in both treatment-naïve and -experienced patients and share many of the same resistance

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mutations, hence establishing the problem of cross-resistance among these drugs (12–15).

In contrast, no resistance mutations have yet been associated with DTG in drug-naïve patients, after >5 years of experience in clinical trials (7, 13). Tissue culture selection studies identified two DTG mutational pathways for resistance, initiated by either the R263K substitution in subtype B and circulating recombinant form AG (CRF02_AG) or the G118R substitution in subtype C and CRF02_AG (16). We showed that R263K and a subsequent H51Y substitution conferred DTG resistance but that the addition of a secondary H51Y mutation caused an additional drop in viral fitness below that conferred by R263K alone (17). This is in contrast to the usual situation whereby secondary resistance mutations commonly play a compensatory role in regard to viral fitness while also increasing levels of drug resistance. The R263K mutation has also been identified in at least two ARV-experienced patients who were treated with a DTG-containing regimen (18). Our results also showed that the G118R substitution can confer resistance against DTG in subtype B integrase (19). The G118R substitution alone and in conjunction with the E138K substitution was previously identified in subtype C but not in subtype B integrase in drug selections in culture with an investigational INSTI, MK-2048 (20). Similarly to DTG, integrase complex-drug binding studies showed that MK-2048 remained bound to the preintegration complex for far longer times than either RAL or EVG, helping to explain its superior profile in regard to the development of resistance in culture (21). Although the clinical development of MK-2048 has not continued beyond phase IIb clinical trials for reasons of poor pharmacokinetics, the selection of G118R by both DTG and MK-2048 as well as the increased susceptibility of the R263K-containing integrase protein to MK-2048 (16) point to intersections and similarities between the resistance profiles of these two compounds.

The purpose of this study was to gain a mechanistic understanding, through cell culture and biochemical and structural analyses, of the differential selection of G118R and its associated substitutions H51Y and E138K in HIV subtypes B and C as well as in CRF02_AG and to evaluate the impact of these substitutions on integrase activity.

MATERIALS AND METHODS

Cells and antiviral compounds. *Escherichia coli* strain XL10-Gold [Tet^r Δ(*mcrA*)183 Δ(*mcrCB*-*hsdSMR*-*mrr*)173 *endA1 supE44 thi-1 recA1 gyrA96 relA1 lac* Hte [F' *proAB lacI*^qΔM15 Tn10(Tet^r) Amy Cam^r] (Stratagene)] ultracompetent cells were used for plasmid production. The integrase inhibitor drugs RAL and MK-2048 were kindly provided by Merck Inc., while EVG and DTG were kindly provided by Gilead Sciences and GlaxoSmithKline, respectively. Prior to use, compounds were solubilized in dimethyl sulfoxide (DMSO) and stored at −20°C. All reagents used were enzyme grade and of high purity.

Plasmids and site-directed mutagenesis. The generation of a pET15b expression plasmid containing soluble HIV subtype B and C integrases was reported previously (16, 22). In order to construct a pET-15b expression plasmid containing the soluble HIV circulating recombinant form (CRF02_AG), the CRF02_AG integrase coding sequence was amplified from p97AG proviral plasmid DNA (GenBank accession number AB052867.1) by PCR using the following primers: p97-INT-FWD (5'-GCCAGGATCCTTTTGTAGTGGCATAGATAAAGCCCAAGAAG-3') and p97-INT-RVS (5'-GTTATCTAGATTAATCCTCATCTGTCTA TCTGCCACACAATC-3'). Amplification products were purified by using the QIAquick PCR purification kit (Qiagen, Toronto, ON, Canada), digested by using the BamHI and XbaI restriction enzymes (NEB, Whitby,

ON, Canada), and ligated inside plasmid pET15b by using T4 DNA ligase (NEB). The resulting plasmid, pET15b-INAG, was verified by sequencing.

PCR-mediated site-directed mutagenesis performed on this plasmid yielded plasmid DNA coding for the G118R mutation, in isolation or together with either of two additional mutations, H51Y and E138K. The following primers were used for mutagenesis: G118R_{AG}-sense (CCAGTGAAAGTGA TACACACAGACAATCGCAGAAATTTCCACC), G118R_{AG}-antisense (GGTGAAATTTCTGCGATTGTCTGTGTGTATCACTTTCCAC TGG), H51Y_{AG}-sense (GCTAAAAGGGGAAGCCATATATGGACAAGTA GACTGT), H51Y_{AG}-antisense (ACAGTCTACTTGTCCATATATGGCTT CCCCTTTTAGC), E138K_{AG}-sense (TTGGTGGACAAATGTTACACAAA AATTGGAATTCCTACAATCC), E138K_{AG}-antisense (GGATTGTAGG GAATTCCAAATTTTGTGTAACATTGTCCACCAA), G118R_C-sense (GGCCAGTCAAAGTAATACATACAGACAATCGTAGTAATTTCA CCAG), G118R_C-antisense (CTGGTGAATTAACGATTGTCTGTAT GTATTACTTTGACTGGCC), H51Y_C-sense (CAAAAGGGGAAGCCATG TATGGACAAGTAGACTGT), H51Y_C-antisense (ACAGTCTACTTGTCC ATACATGGCTTCCCCTTTTG), E138K_C-sense (GGGCAGGTATCCAAC AGAAATTGGGATTCCCTAC), and E138K_C-antisense (GTAGGGA ATCCCAAATTTCTGTTGGATACCTGCC). Successful mutagenesis was confirmed by sequencing (Genome Quebec).

Protein expression, purification, and quantification. Plasmids bearing either a wild-type (WT) or an appropriately mutated IN coding sequence were transformed into BL21(DE3) Gold [F[−] *ompT hsdS_B* (*r_B*[−] *m_B*[−]) *dcm* Tet^r *gal* λ(DE3) *endA* Hte (Stratagene)] cells for protein expression. Luria-Bertani (LB) broth (Multicell), prepared in MilliQ water and supplemented with 100 μg/ml ampicillin, was used for all bacterial growth. Expression and purification of integrase recombinant proteins were performed as previously described for hexahistidine-tagged integrase (16). Fractions containing purified integrase, as judged by SDS-PAGE, were dialyzed into storage buffer (20 mM HEPES, 1 M NaCl, 1 mM EDTA, 5 mM dithiothreitol [DTT], 10% glycerol [pH 7.5]). Protein concentrations were measured by using a calculated molar extinction coefficient of 50,420 M^{−1} cm^{−1}; of note, the molar extinction coefficients of all integrase proteins were the same and in congruence with that calculated for subtype B integrase (16). Protein aliquots were kept for several months at −80°C without a significant loss of enzymatic activity.

DNA substrates for *in vitro* assays. All oligonucleotide substrates were purchased from Integrated DNA Technologies (IDT, Coralville, IA, USA). The following oligonucleotides were used for strand transfer assays: preprocessed donor LTR DNA sense primer A, tagged at the 5' end by a 12-carbon linker to a reactive amino group (AmMC12) (5'-AmMC12-A CCCTTTAGTCAGTGTGGAAAATCTCTAGCA-3'), and antisense primer B (5'-ACTGCTAGAGATTTTCCACACTGACTAAAAG-3') and biotinylated (Bio) target DNA sense primer C (5'-TGACCAAGGGCTAA TTCACT-3' Bio) and antisense primer D (5'-AGTGAATTAGCCCTTGG TCA-3' Bio). For 3' processing assays, primer B was used together with LTR 3' sense oligonucleotide E, which is 5' modified with an AmMC12 tag and modified at the 3' end with a biotin tag attached via a triethylene glycol spacer (BioTEG) (5'-AmMC12-ACCCTTTTAGTCAGTGTGGAA AATCTCTAGCAGT-BioTEG-3').

The key features of these oligonucleotides were previously described (16, 23). Functional DNA duplexes were made by combining equimolar quantities of sense and antisense primers to appropriate concentrations in low-chelate Tris-EDTA (TE) buffer (10 mM Tris, 0.1 mM EDTA [pH 7.8]). The combined primers were heated for 10 min at 95°C and annealed by slow cooling to room temperature over a period of 4 h. Duplexes were stored at −20°C for several months without a loss of integrity.

Preparation of preprocessed LTR-coated plates for strand transfer activity. Eighty microliters of preprocessed viral LTR-mimic (donor DNA) duplex A/B diluted to 300 nM (except as dictated by experimental design) in phosphate-buffered saline (PBS) (pH 7.4) (Bioshop) was covalently linked to Costar DNA-Bind 96-well plates (catalog number 2499; Corning) by overnight incubation at 4°C. The plates were blocked with 0.5% bovine serum albumin (BSA) in blocking buffer (20 mM Tris [pH

7.8], 150 mM NaCl) and stored in blocking buffer for several weeks without a detectable loss of integrity. Before use, the plates were washed twice with PBS (pH 7.4) and then assay buffer {50 mM morpholinepropanesulfonic acid (MOPS) (pH 6.8), 50 μ g/ml BSA, 50 mM NaCl, 0.15% 3-[(3-cholamidopropyl)-dimethylammonio]-1-propanesulfonate (CHAPS), 30 mM $\text{MnCl}_2/\text{MgCl}_2$ }, except as dictated by experimental design.

Protein calibration for strand transfer activity. In order to better elucidate the impact of resistance substitutions, the strand transfer and 3' processing steps of the HIV integration reactions were decoupled, as previously described (16, 19). The optimum concentration of protein for use in strand transfer experiments was determined by titration, as previously described (16, 19). Briefly, purified integrase proteins, appropriately diluted in reaction buffer (50 mM MOPS [pH 6.8], 50 mg/ml BSA, 50 mM NaCl, 30 mM MnCl_2 , 0.015% CHAPS, 5 mM DTT), were added to preprocessed LTR-bound plates as described above. This step was followed by a 30-min incubation at room temperature to allow for the assembly of integrase onto the preprocessed LTR duplexes. Various concentrations of biotinylated target DNA duplex C/D (0 to 240 nM) were added, followed by a 1-h incubation at 37°C for the strand transfer reaction to occur. The plates were then rinsed three times with wash buffer (50 mM Tris [pH 7.5], 150 mM NaCl, 0.05% Tween 20, 0.2 mg/ml BSA) and incubated with 150 μ l of Eu-labeled streptavidin (PerkinElmer) diluted to 0.025 μ g/ml in wash buffer in the presence of 50 μ M diethylenetriamine pentaacetate (DTPA) (Sigma). After additional rinses with wash buffer, 100 μ l of Wallac enhancement solution (PerkinElmer) was added. The amount of target DNA covalently linked to the LTR was then measured by quantifying the amount of bound Eu-labeled streptavidin. The low pH of the Wallac solution caused the ionization of Eu to Eu^{3+} . Excitation of Eu^{3+} ions by incident wavelength at 355 nm resulted in time-resolved emission of fluorescence (TRF), which was measured at 612 nm on a FLUOStar Optima multilabel plate reader (BMG Labtech).

Protein concentrations that yielded the highest activity, as measured by relative fluorescence units (RFU), were chosen for subsequent experiments.

Strand transfer activity assay. Michaelis-Menten enzyme analysis has previously been used to determine enzyme parameters of HIV integrase (24). While 3' processing has been conclusively shown to follow non-Michaelis-Menten kinetics (25, 26), there has been no similar study of HIV integrase strand transfer. In order to obtain relative activity parameters for wild-type and variant integrase proteins, strand transfer reactions were carried out based on the same design as that for the protein calibration assay described above. The major deviation was that an effective LTR concentration of 160 nM and an effective integrase concentration of 300 nM were used for these assays, while the target DNA concentration varied from 0 to 128 nM. All other experimental procedures were performed as described above. The means of data from at least three independent experiments, each performed in triplicate for the wild-type integrase and each variant integrase protein, were calculated. The values of enzyme activity in RFU (A) and target DNA concentration (S) were fit by nonlinear regression using GraphPad Prism V5 (GraphPad Software, San Diego, CA, USA) to the following the Michaelis-Menten derivative equations: $A = A_{\text{max}}' \cdot [S]/([S] + K_m')$ and $K_m' = A_{\text{max}}' \cdot [S]/A - [S]$.

In this case, the pseudo-Michaelis constant (K_m') reflects the apparent affinity of the enzyme for the target DNA substrate ($[S]$), and the maximum activity (A_{max}') reflects the maximum activity obtainable with same concentration of the protein in a similar assay, regardless of the substrate concentration. To determine overall enzyme efficiency, we calculated the ratio of A_{max}' to K_m' , as previously described (19), which is presented as a percentage of wild-type values.

3' processing activity assay. The 3' processing activity of the purified recombinant integrase proteins was determined as previously described (23). Briefly, 3'-biotinylated LTR duplex G/B was covalently linked at various concentrations (effective concentration, 0 to 160 nM) to Costar DNA-Bind plates under conditions similar to those described above for strand transfer assays. To initiate 3' processing, purified integrase protein

(300 nM) diluted in reaction buffer was added, and the plates were incubated at 37°C. Negative protein control wells had only reaction buffer added. After 2 h of incubation, the reaction was within the linear phase (23), and the plates were quickly washed three times with wash buffer to remove all traces of protein and cleaved 3'-GT dinucleotide. All subsequent steps of the assay were the same as those described above for the strand transfer assay, with the exception of data analysis. For each plate and concentration of LTR, four negative protein controls contained unprocessed LTR, and the average signal from these controls represented the maximum possible signal ($3'\text{-OH}_{\text{max}}$). Thus, the RFU representing actual 3' processing activity ($3'\text{-OH}_{\text{actual}}$) was determined by the following calculation: $(3'\text{-OH}_{\text{actual}}) = (3'\text{-OH}_{\text{max}}) - (3'\text{-OH}_{\text{obs}})$.

The calculated 3' processing readings ($n \geq 3$) were then processed by using GraphPad Prism to yield 3' processing enzyme parameters, as previously described (23). The data within the linear phase of the 3' processing reaction were used to calculate kinetic parameters by fitting the data to the Michaelis-Menten equation (24). As such, relative enzyme efficiency calculations of 3' processing activity were calculated as described above for strand transfer.

Confirmation of the impact of G118R and E138K on subtype C integrase inhibition by MK-2048. Since the selection of the G118R substitution in cell culture was initially driven by MK-2048 drug pressure (20) in a subtype C HIV-1 backbone, we performed preliminary confirmation that the G118R and the associated E138K substitutions caused resistance to MK-2048. The dose-dependent susceptibility of the subtype C WT, G118R, E138K, and G118R/E138K proteins to MK-2048 was tested. Briefly, strand transfer assays were performed as described above, with slight modifications. Following incubation of subtype C WT, G118R, E138K, and G118R/E138K proteins with LTR DNA, dose-ranging amounts of MK-2048 (20 pM to 20 μ M) dissolved in compound dilution buffer were added to each well (25 μ l each). After an incubation period of 10 min at room temperature, 25 μ l of target DNA (6 nM) was added to each well, and the strand transfer reaction was allowed to proceed as described above. All subsequent steps were performed as described above for the strand transfer assay. Data obtained were normalized and fit to the following log (inhibitor)-versus-normalized response equation with GraphPad Prism to yield estimates of the 50% inhibitory concentration (IC_{50}) of the drug for the different proteins: $A = 100/[1 + 10^{([I] - \log \text{IC}_{50})}]$.

In the above-described equation, A represents residual strand transfer activity at a given drug concentration, $[I]$. Each experiment was done in triplicate. The fold change (FC) in the susceptibility of each variant to MK-2048 was calculated by dividing the IC_{50} for that variant by that of the WT subtype C integrase protein tested at the same time. Mean FC susceptibility data from at least three independent experiments were graphed by using GraphPad Prism.

Competitive inhibition of strand transfer by DTG, RAL, and EVG. The susceptibilities of our purified integrase proteins to INSTIs were tested in competitive inhibition assays using DTG, RAL, and EVG. Drug stock solutions were prepared as a 6 $\times$ working solution (6,000 nM) and subsequently serially diluted 4-fold into compound dilution buffer (assay buffer without cations and containing 10% DMSO). The assay mixture concentrations varied from 0.2 to 1,000 nM; inhibition assays were performed in the presence of various target DNA concentrations (0 to 128 nM). Briefly, preprocessed LTR-coated plates (effective LTR concentration of 160 nM) were prepared as described above. Purified integrase proteins (effective concentration of 300 nM), in assay buffer supplemented with 5 mM DTT, were added to each well, followed by a 30-min incubation at room temperature to allow for the assembly of integrase onto the preprocessed LTR duplexes. Twenty-five microliters of the appropriately diluted compound was added to each well, followed immediately by the addition of 25 μ l of the appropriately diluted biotinylated target DNA duplex. The plates were then incubated for 1 h at 37°C, followed by the post-strand transfer steps described above. The data from at least three independent competition assays for each protein were fit to

TABLE 1 Selection of G118R and associated substitutions in cell culture by MK-2048 and DTG^a

Subtype	Baseline polymorphisms	Wk	Acquired mutation(s)	Wk	Acquired mutation(s)	Reference
MK-2048 selections						
B	M154I, V201I	34	None			5
C	V72I, Q95P, T125A	25	G118R	29	G118R, E138K	
DTG selections						
B	M154I, V201I	20	R263K	37	E138E/K, R263K	3
AG	V72I, T125A, V201I	20	R263K	37	H51H/Y, R263K	
AG	V72I, T125A, V201I	20	E69E/K, G118R	37	G118R	
C	V72I, Q95P, T125A	20	G118R	37	H51Y, G118R	
C	V72I, T125A, V201I	20	S153S/T	37	H51Y, G193G/E	

^a Substitutions tested in this work are shown in boldface type.

the competitive inhibition algorithm using GraphPad Prism as follows: $K_{mobs} = K_m' \cdot (1 + [I]/K_i)$ and $A = A_{max}' \cdot [S]/(K_{mobs} + [S])$.

In the above-described equation, K_{mobs} represents the Michaelis constant obtained in the presence of the inhibitor, $[I]$ represents the inhibitor concentration, and K_i is the inhibition constant of the inhibitor for that particular protein. All other abbreviations are defined above. During analysis, the K_m' for each protein was constrained to value previously determined in the strand transfer activity assays described above, while V_{max}' was left unconstrained.

K_i values calculated by this algorithm were transformed to FC values by division of the K_i values for variant proteins by the K_i value for the appropriate wild-type integrase enzyme. FC calculations from multiple inhibition assays were compiled and analyzed by using GraphPad Prism.

Monogram Biosciences PhenoSense replication capacity and phenotyping assays. The HIV replication capacity and susceptibilities to DTG, RAL, and EVG were measured as previously described (27). Briefly, murine leukemia virus envelope-pseudotyped HIVs bearing WT, H51Y, G118R, E138K, or G118R/E138K integrases and a luciferase reporter gene were used to inoculate human embryonic kidney HEK-293 cells. The resultant luciferase activity was used to calculate changes in HIV replication capacity relative to that of a wild-type reference strain. Drug susceptibility was expressed as a fold change in the IC_{50} relative to that of the wild type.

Data processing. Data for all cell-free experiments presented here, except where otherwise indicated, were the results of at least 3 independent sets of experiments. When relevant, the statistical significance of differences between data sets for two or more integrase proteins was determined by using a one-sample two-tailed t test with GraphPad Prism. Probability values of ≤ 0.05 ($P \leq 0.05$) were indicative of statistically significant differences between the different proteins tested.

Homology modeling and active-site analysis. Homology models of integrase proteins were generated by using the ProtMod server (<http://ffas.burnham.org/>), based on lead subtype B model templates that were initially generated by using the I-TASSER protein modeling server (28). The prototype foamy virus (PFV) structure reported under PDB accession number 4E7K (29) was utilized as a lead template to generate a subtype B WT monomeric model for the formation of the target capture complex (TCC). The PFV integrase structure reported under PDB accession number 3S3M was used as a lead template to create the I-TASSER model for DTG-bound HIV integrase. The ProtMod server was used to minimize stochastic errors between the models and to remove any sampling errors introduced by the multitemplate modeling method (30) of I-TASSER. Briefly, single-template (subtype B WT) and single-query (each integrase sequence, including that of subtype B WT integrase) alignments were performed by using the alignment program SCWRL (31). The program Modeler (27) was then used to create monomeric homology models of each integrase based on the SCWRL sequence alignment and the subtype B WT I-TASSER structure. Model quality was assessed based on the root mean square deviation (RMSD) of the global homology structure from the PFV lead template by using the RCSB PDB Protein

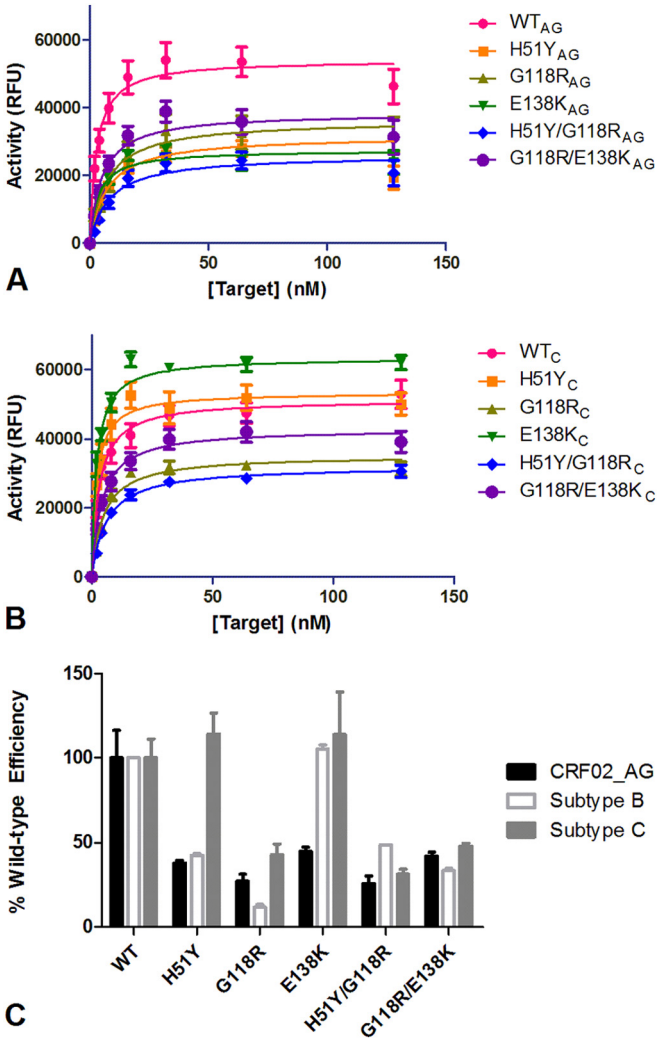


FIG 1 Comparative strand transfer activities of purified HIV-1 WT integrase and variant integrase proteins of CRF02_AG, subtype C, and subtype B origins. (A) Target DNA saturation (0 to 128 nM) plots showing the activity of CRF02_AG (AG) proteins in the presence of fixed protein and LTR concentrations (300 nM and 160 nM, respectively). (B) Target DNA saturation plots showing the activity of subtype C proteins in the presence of fixed protein and LTR concentrations (300 nM and 160 nM, respectively). (C) Comparison of strand transfer reaction efficiencies for CRF02_AG, subtype B, and subtype C integrase proteins. All data presented reflect at least three independent experiments, each performed in duplicate or triplicate.

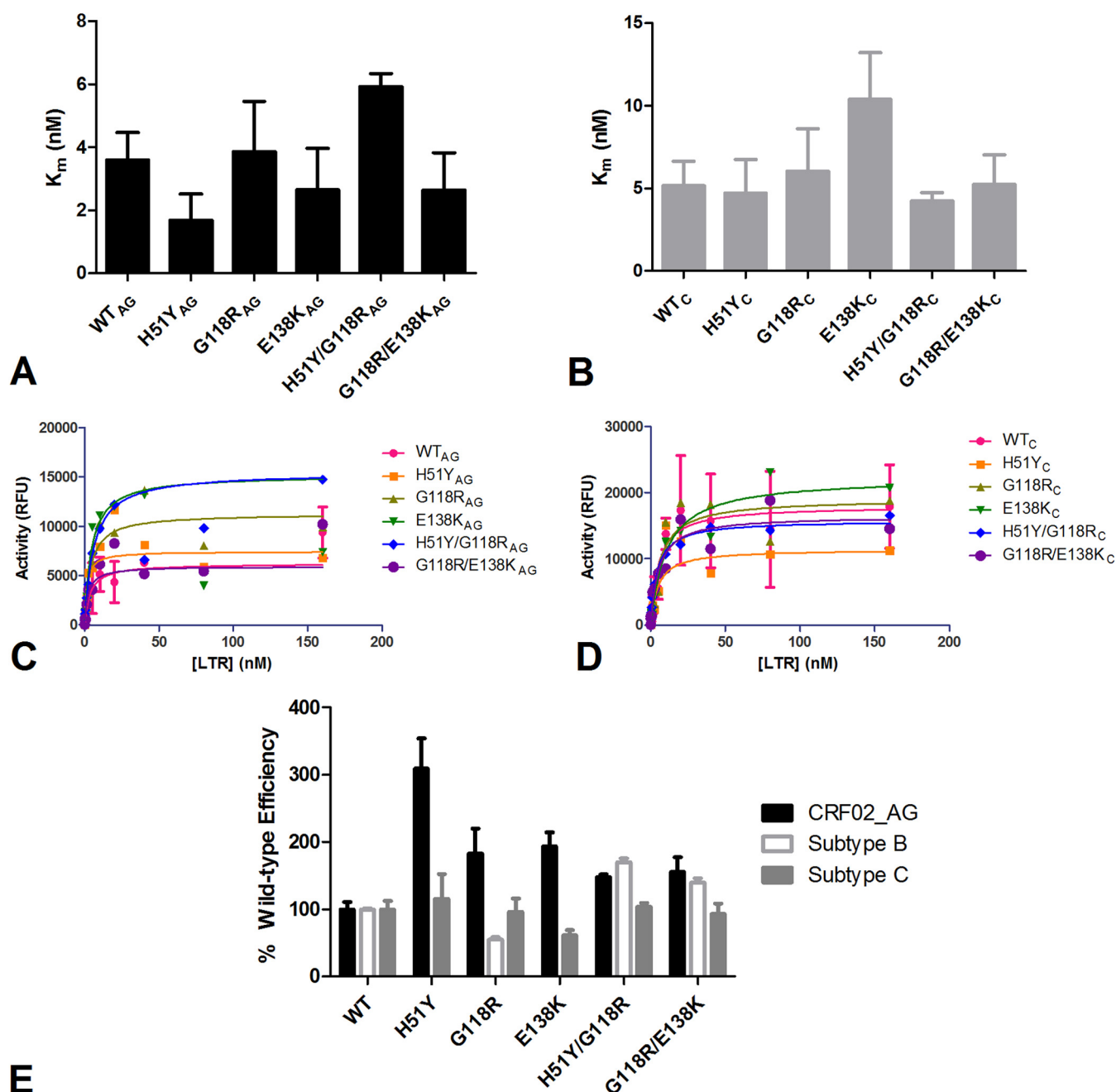


FIG 2 Comparative 3' processing activities of purified HIV-1 WT and variant integrase proteins of CRF02_AG, subtype C, and subtype B origins. (A and B) Effect of amino acid substitution on functional binding (K_m) of the viral LTR mimic by CRF02_AG integrase (A) or subtype C integrase (B). (C and D) LTR DNA saturation plots showing the activity of CRF02_AG (C) or subtype C (D) proteins in the presence of a fixed protein concentration (300 nM). (E) Comparison of 3' processing reaction efficiencies for subtype B, subtype C, and CRF02_AG integrase proteins. All data presented reflect at least three independent experiments, each performed in duplicate or triplicate.

Comparison Tool (32). All structures were also verified as having >89% of residues in allowed and favorable orientations by Ramachandran plot analysis (33, 34). Superimposition of the HIV-1 homology models with the reported cocrystal structure of PFV with DTG (PDB accession number 3S3M) provided insights into the mechanism of resistance caused by the G118R substitution in regard to this and other INSTIs. DNA interaction hints were obtained by overlaying the HIV-1 homology models with the PFV crystal structure reported under PDB accession number 4E7K, representing the TCC (29). The molecular

visualization program PyMOL (PyMOL Molecular Graphics System, version 1.3; Schrödinger, LLC [http://PyMOL.org/]) was used for structural visualization and image processing.

RESULTS

Generation of recombinant HIV subtype C and CRF02_AG integrase enzymes. Two resistance selection studies performed in our laboratory with the investigative drugs MK-2048 and

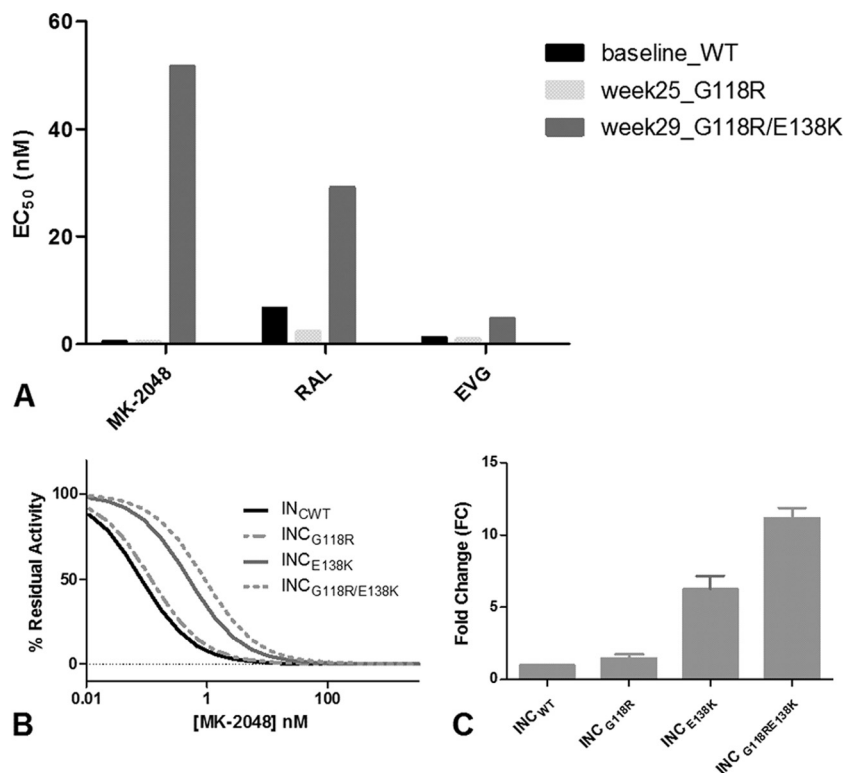


FIG 3 Confirmation of the role of the G118R and E138K substitutions in conferring resistance to MK-2048 in subtype C integrase. (A) Susceptibility of MK-2048-selected subtype C variant viruses to MK-2048, RAL, and EVG. (B) Dose-dependent inhibition of the strand transfer reaction of subtype C WT and G118R-, E138K-, and G118R/E138K-containing integrase proteins (INC) by the INSTI MK-2048 (0.173 nM to 5,000 nM). (C) Calculation of the FCs in IC₅₀ values relative to WT values showing that the G118R/E138K double-variant protein has high-level *in vitro* resistance to MK-2048. The data presented here are the results from at least three independent experiments performed in triplicate.

S/GSK139572 (DTG) yielded the G118R mutation in subtype C (20) and CRF02_AG (16) integrases, respectively. Key amino acid changes observed in these selection studies are presented in Table 1. Recombinant enzymes, either of the wild-type sequence or site mutated to contain H51Y, G118R, or E138K, H51Y/G118R, and G118R/E138K, were expressed and purified as previously described (16, 17, 19). As in previous studies, protein calibration confirmed that maximum protein activity occurred with 300 nM protein for all subtypes (not shown). Therefore, subsequent experiments utilized 300 nM integrase protein. When relevant, we compared our results with those obtained by using the subtype B protein.

Effect of G118R on integrase strand transfer efficiency alone or in combination with H51Y or E138K. Given that G118R and mutations associated with it arose during serial passage with several INSTIs, we expected these mutations to impact the strand transfer reaction. In CRF02_AG integrase, the presence of any of the H51Y, G118R, and E138K substitutions significantly reduced the enzyme parameters of the strand transfer reaction (Fig. 1A) and led to reductions in enzyme efficiency of >50% (Fig. 1C). H51Y caused a significant loss of enzyme efficiency in CRF02_AG and subtype B but not in subtype C integrase (Fig. 1B and C). The E138K single mutation was also not associated with a drop in enzyme efficiency in subtype B and C integrase proteins but caused a >55% loss of activity when present in CRF02_AG integrase. Of note, G118R caused significant diminutions in activity for all subtypes treated, i.e., subtype C (43% of WT activity),

CRF02_AG, (27%), and subtype B (12%) (Fig. 1C). The addition of H51Y and E138K to G118R partially restored the strand transfer efficiency of subtype B integrase but less so with the subtype C or CRF02_AG integrase enzyme (Fig. 1C).

Impact of the G118R, H51Y, and E138K substitutions on 3' processing ability. To fully assess the impact of the G118R, H51Y, and E138K substitutions on integration, it is essential that the 3' processing ability of the various integrase proteins be evaluated. Accordingly, cell-free 3' processing experiments were performed by using dose-ranging levels of the viral LTR mimic, as previously described (Fig. 2) (19). The LTR DNA binding ability of the recombinant proteins, as indirectly inferred from K_m' values, was not significantly different from that of the WT, except for H51Y for both non-B subtypes and E138K for subtype C (Fig. 2A and B). H51Y resulted in markedly tighter binding to the LTR for both the CRF02_AG and subtype C integrase proteins, while the opposite result was obtained for subtype C integrase containing the E138K substitution (Fig. 2A and B). G118R alone or in combination with either H51Y or E138K did not cause a significant change in LTR binding for either subtype. However, the doubly substituted H51Y/G118R CRF02_AG protein showed lower LTR DNA binding than that of the WT.

The impacts of these mutations on overall 3' processing activity were markedly different among the subtypes tested (Fig. 2C to E). All CRF02_AG variant proteins exhibited significantly higher 3' processing activity (Fig. 2C) and, thus, higher enzyme efficiency (Fig. 2E) than did the WT protein. In the subtype C protein, the

TABLE 2 Enzymatic and virological parameters of variant CRF02_AG and subtype B and C HIV-1 integrase proteins and viruses^a

HIV-1 subtype	Integrase phenotype	Mean enzyme efficiency (% of WT) $\pm$ SD			Mean INSTI susceptibility (fold change relative to WT) $\pm$ SD					
		3' processing	Strand transfer	Viral replication capacity (% of WT)	DTG		RAL		EVG	
					Cell free	Cell culture	Cell free	Cell culture	Cell free	Cell culture
CRF02_AG	WT	100 $\pm$ 10.9	100 $\pm$ 16.4	100	1.00	1.0	1.00	1.0	1.00	1.0
	H51Y	309 $\pm$ 45.1	38.0 $\pm$ 1.4	69	1.47 $\pm$ 0.13	1.4	2.68 $\pm$ 0.08	1.0	0.631 $\pm$ 0.009	2.1
	G118R	182 $\pm$ 37.4	27.0 $\pm$ 4.4	8	3.17 $\pm$ 0.67	15.5	8.48 $\pm$ 2.05	17.2	2.58 $\pm$ 0.77	7.7
	E138K	193 $\pm$ 20.4	44.8 $\pm$ 2.5	91	2.61 $\pm$ 0.18	0.8	1.14 $\pm$ 0.23	0.8	5.87 $\pm$ 1.18	1.0
	H51Y/G118R	147 $\pm$ 4.1	25.9 $\pm$ 4.4	ND	2.24 $\pm$ 0.26	ND	0.595 $\pm$ 0.018	ND	0.617 $\pm$ 0.048	ND
	G118R/E138K	155 $\pm$ 21.4	41.9 $\pm$ 2.4	32	1.33 $\pm$ 0.35	12.7	3.33 $\pm$ 0.78	20.2	1.39 $\pm$ 0.28	5.2
B	WT	100 $\pm$ 1.5	100 $\pm$ 0.4	100	1.00	1.0	1.00	1.0	1.00	1.0
	H51Y	ND	42.5 $\pm$ 1.2	89	3.75 $\pm$ 0.82	1.4	1.43 $\pm$ 0.53	1.2	3.53 $\pm$ 2.49	2.0
	G118R	54.7 $\pm$ 3.8	11.8 $\pm$ 1.7	30	3.11 $\pm$ 0.58	8.2	3.68 $\pm$ 0.75	10.6	4.62 $\pm$ 2.35	5.5
	E138K	ND	105 $\pm$ 2.2	77	1.83 $\pm$ 0.13	0.8	3.84 $\pm$ 0.37	1.0	1.30 $\pm$ 0.03	0.8
	H51Y/G118R	170 $\pm$ 5.5	48.4 $\pm$ 0.04	ND	3.43 $\pm$ 0.97	ND	2.78 $\pm$ 0.65	ND	3.18 $\pm$ 0.45	ND
	G118R/E138K	140 $\pm$ 6.6	33.4 $\pm$ 1.46	43	0.644 $\pm$ 0.047	8.0	2.18 $\pm$ 0.13	14.1	0.884 $\pm$ 0.029	4.8
C	WT	100 $\pm$ 12.5	100 $\pm$ 11.2	ND	1.00	ND	1.00	1.0 ^b	1.00	1.0 ^b
	H51Y	116 $\pm$ 36.8	114 $\pm$ 12.6	ND	2.51 $\pm$ 1.22	ND	5.80 $\pm$ 2.28	ND	2.77 $\pm$ 0.57	ND
	G118R	96.4 $\pm$ 19.9	42.6 $\pm$ 6.6	ND	2.31 $\pm$ 0.60	ND	1.93 $\pm$ 0.64	0.3 ^b	1.93 $\pm$ 0.58	0.78 ^b
	E138K	61.4 $\pm$ 7.3	114 $\pm$ 12.6	ND	1.47 $\pm$ 0.26	ND	1.30 $\pm$ 0.31	ND	1.07 $\pm$ 0.26	ND
	H51Y/G118R	103 $\pm$ 5.5	31.3 $\pm$ 3.0	ND	3.78 $\pm$ 1.21	ND	1.51 $\pm$ 0.26	ND	1.61 $\pm$ 0.29	ND
	G118R/E138K	93.1 $\pm$ 15.3	47.9 $\pm$ 1.6	ND	2.93 $\pm$ 1.00	ND	2.20 $\pm$ 1.01	4.4 ^b	0.876 $\pm$ 0.221	4.1 ^b

^a ND, not determined.^b Phenotyping studies performed on infectious viruses selected during MK-2048 selections in cord blood mononuclear cells (CBMCs).

presence of the E138K substitution resulted in a 38% reduction in the efficiency of 3' processing, while the H51Y and E138K substitutions were without effect. A diminished 3' processing ability was observed for the G118R integrase protein of subtype B but not of subtype C or CRF02_AG.

Effect of G118R and E138K on subtype C integrase protein resistance to MK-2048. The G118R substitution was selected in cell culture by MK-2048 in a subtype C virus (20). Despite being selected by subtype C isolate 4742, the addition of E138K (within 4 weeks) was necessary to engender significant resistance of the virus to MK-2048, with a fold change (FC) of the 50% effective concentration (EC₅₀) of $\sim$ 139 (Fig. 3A), and lower levels of cross-resistance to RAL (FC, $\sim$ 4.4) and EVG (FC, $\sim$ 4.1). We also studied the susceptibility of subtype C WT integrase or integrase containing the G118R, E138K, or G118R/E138K substitution to MK-2048 (Fig. 3 and Table 2). As previously reported for subtype B in cell culture (20), the G118R substitution resulted in minimal resistance to MK-2048 (FC, $\sim$ 1.5) (Fig. 3A and B), and this level of resistance was significantly increased by the presence of the E138K mutation (FC, $\sim$ 11). Of note, the FC in resistance to MK-2048 reported for the clonal G118R/E138K subtype C virus is $\sim$ 100-fold higher than the one previously reported for a subtype B virus (20). This result may partially account for the emergence of G118R and the additional selection of E138K in subtype C viruses (20).

Effect of G118R on susceptibility to DTG, RAL, and EVG. We next tested the susceptibilities of subtype C and CRF02_AG integrase proteins, containing G118R alone or in combination with H51Y or E138K, as well as those containing H51Y or E138K alone, to DTG, RAL, and EVG (Fig. 4 and Table 2). The G118R substitution in all three subtypes resulted in statistically similar levels of

resistance to DTG (Fig. 4A and Table 2) and significant cross-resistance with RAL (Fig. 4B) and EVG (Fig. 4C). However, G118R in subtypes B and C resulted in only low-level resistance to RAL, whereas an 8-fold change in susceptibility to RAL was noted for CRF02_AG, consistent with clinical results (35); no statistical differences between subtypes were observed in regard to resistance to EVG. The impacts of the two secondary mutations H51Y and E138K on G118R-bearing protein differed between subtypes. The H51Y/G118R combination in subtype C, but not in CRF02_AG or subtype B, resulted in slightly elevated levels of resistance to DTG compared to that with G118R alone (Fig. 4A and Table 2). Consistent with this, G118R/H51Y was selected by DTG in subtype C (Table 1). Although the H51Y/G118R variant integrase enzyme showed low levels of cross-resistance to both RAL and EVG in subtype B (FC, $\sim$ 3), susceptibility levels were comparable to those of the WT for the subtype C protein (FC, <2). H51Y/G118R, in the CRF02_AG protein, displayed greater susceptibility to both RAL and EVG than the WT (FC, $\sim$ 0.6) (Fig. 4B and C and Table 2). Variant enzymes bearing E138K showed the greatest subtype-dependent variability; compared to the WT, the G118R/E138K variant integrase proteins of subtype B, subtype C, and CRF02_AG showed decreased susceptibility, increased susceptibility, and similar susceptibility to DTG, respectively (Fig. 4A and Table 2). However, all 3 subtypes exhibited similar levels of cross-resistance to RAL (FC, $\sim$ 2) (Fig. 4B and Table 2) and WT levels of susceptibility to EVG (Fig. 4C and Table 2). The E138K variant in the three subtypes exhibited low-level resistance to DTG (FC, $\sim$ 1.5 to 2) (Fig. 4A and Table 2), low-level cross-resistance to RAL in subtype B (Fig. 4B and Table 2), and high-level cross-resistance to EVG in CRF02_AG (Fig. 4C and Table 2).

Clonal WT, H51Y, G118R, E138K, or G118R/E138K viruses

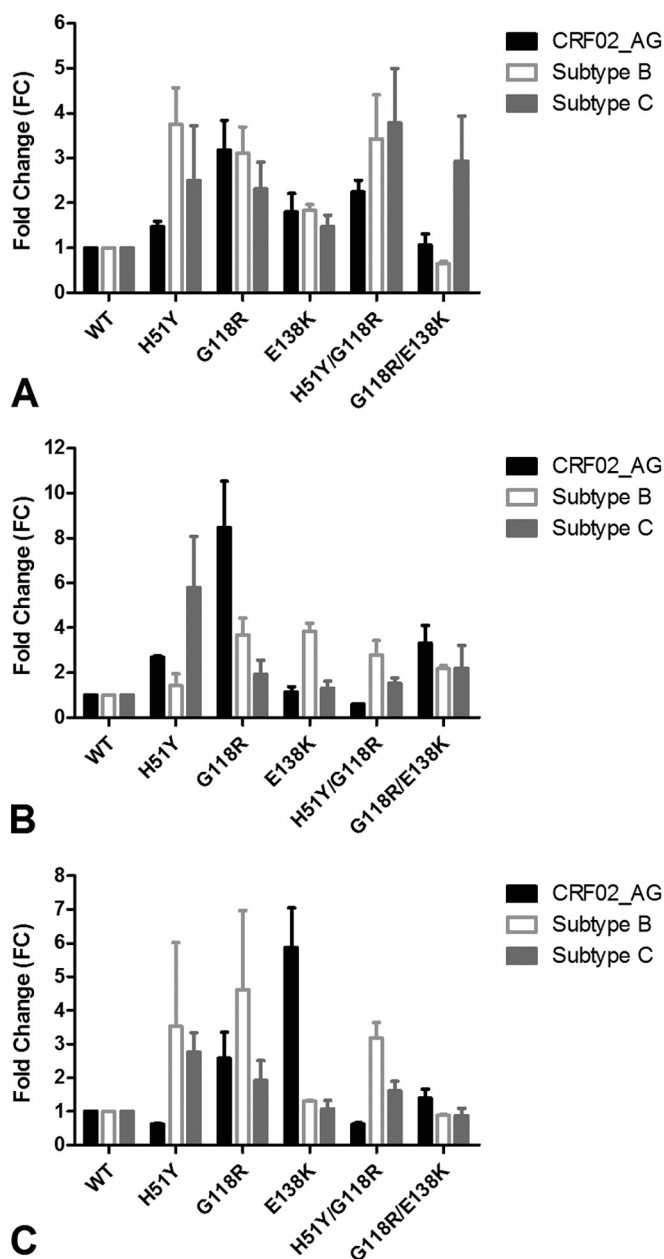


FIG 4 Subtype-specific susceptibility of WT and variant integrase proteins to clinically relevant INSTIs. (A) DTG; (B) RAL; (C) EVG. K_i values were derived by performing strand transfer assays with variable drug (0.2 nM to 1,000 nM) and variable target DNA (0 nM to 128 nM) concentrations in the presence of fixed concentrations of LTR (160 nM) and integrase protein (300 nM). Data were fit by nonlinear regression analysis using GraphPad Prism and by a competitive inhibition equation, as detailed in Materials and Methods. FC values were calculated for each experiment by dividing the observed K_i value of each variant for a particular INSTI by that observed for the WT with the same INSTI. For each subtype, FC calculations from at least three individual experiments, analyzed by using column statistics, are presented.

were phenotyped by Monogram Biosciences in the subtype B and CRF02_AG backgrounds (Tables 2 and 3). Difficulties in generating clonal variants of subtype C precluded its similar analysis. The phenotyping data imply that G118R alone conferred significant yet low-level resistance to DTG in both subtype B and CRF02_AG

but with a significant loss of viral replication capacity (Table 2). G118R also caused significant cross-resistance to EVG and RAL, particularly to RAL in CRF02_A/G (35). Where possible, when the calculated resistance levels and viral replication in cell culture were compared to the calculated resistance and integration activity from biochemical assays (Table 3), there was good agreement between the two methods for most variants.

Amino acid sequences differ at key positions between the three subtypes. We performed multiple-sequence alignments using ClustalW v1.8 software (36) to try to explain the subtype-specific differences in these experimental data. Integrase amino acid sequences from subtype B (pNL4_3), subtype C (pINDieC), and CRF02_AG (p97) were aligned (Fig. 5). The three integrase amino acid sequences share >93% sequence identity and differed at only at a few positions, mostly through conservative polymorphisms. Interestingly, several polymorphic positions appear to be close to second-generation INSTI resistance-associated positions. As an example, the amino acid at position 50 in treatment-naïve patients is mostly M in subtype B (37), with polymorphisms in other subtypes. The M50I substitution appeared as a secondary mutation to R263K during passage of subtype B HIV in the presence of DTG (16) and increased the levels of resistance to DTG, together with R263K, without affecting the viral fitness cost imposed by R263K (37). Position 50 is also only one residue downstream of the DTG resistance-associated substitution H51Y, which appears to be a secondary substitution for both G118R and R263K that increased the levels of resistance to both DTG and EVG while also increasing the fitness cost imposed by R263K in subtype B HIV (17).

Another key polymorphic position at residue 91 is naturally A in both subtypes B and C and is E in CRF02_AG. CRF02_AG integrase possesses an R at position 119, while subtypes B and C have an S at this position, and all three subtypes have different residues at positions 133 to 136. Of note, positions 140, 143, 148, and 155 (Fig. 5, boxed in blue), which are associated with primary drug resistance to RAL and EVG, all occur within highly invariant motifs. Although the C terminus of integrase is known to be highly polymorphic across retroviral genera, it is highly conserved among the three subtypes studied here.

DISCUSSION

The integrase inhibitor-associated substitution G118R has never been selected in subtype B viruses by any drug, either in the clinic or in a laboratory setting. In contrast, it has been selected in culture in subtype C and CRF02_AG clonal viruses (Table 1) (16, 20) and in the clinic in a CRF02_AG virus (35). We subsequently showed that the presence of this mutation, when introduced into subtype B integrase, caused resistance to DTG, EVG, and RAL (19) at both an integrative cost (19) and a fitness cost to the virus (20). In this study, we investigated the biochemical basis for the differential selection of G118R and its associated H51Y and E138K amino acid substitutions in HIV subtypes B and C as well as CRF02_AG. We evaluated the impact of G118R alone or in concert with either H51Y or E138K on the ability of recombinant integrase proteins from subtypes B, C, and CRF02_AG to perform strand transfer and 3' processing activities, and the individual impacts of H51Y and E138K were determined. We also evaluated the phenotypes of HIV-1 clonal viruses bearing wild-type or variant integrase sequences. After verifying that the sequential selection of G118R and E138K caused resistance to MK-2048, we also

TABLE 3 Comparative analysis of biochemical and virology data

HIV-1 subtype	Integrase phenotype	Extent of INSTI susceptibility (FC relative to WT) ^a						% integration activity ^b	
		DTG		RAL		EVG		EE	RC
		Cell free	Cell culture	Cell free	Cell culture	Cell free	Cell culture		
CRF02_AG	WT	—	—	—	—	—	—	100	100
	H51Y	—	—	++	—	—	+	117	69
	G118R	++	+++	+++	+++	++	++	49.1	8
	E138K	++	—	—	—	+++	—	86.5	91
	H51Y/G118R	+	ND	—	ND	—	ND	38.1	ND
	G118R/E138K	—	+++	++	+++	++	++	64.9	32
B	WT	—	—	—	—	—	—	100	100
	H51Y	++	—	—	—	++	+	ND	89
	G118R	++	++	++	+++	+++	+++	54.7	30
	E138K	+	—	++	—	—	—	ND	77
	H51Y/G118R	++	ND	++	ND	++	ND	82.3	ND
	G118R/E138K	—	++	+	+++	—	+	46.8	43
C	WT	—	ND	—	ND	—	ND	100	ND
	H51Y	++	ND	+++	ND	++	ND	132	ND
	G118R	+	ND	+	ND	+	ND	96.4	ND
	E138K	—	ND	—	ND	—	ND	70.0	ND
	H51Y/G118R	++	ND	+	ND	+	ND	32.2	ND
	G118R/E138K	++	ND	+	ND	—	ND	93.1	ND

^a Cell culture experiments tend to yield higher FC values than do cell-free assays (>3) (16). Susceptibility rankings of FC for cell-free experiments are as follows: —, 0 to <1.5; +, 1.5 to <2.5; ++, 2.5 to <4.0; +++, ≥4.0. In cell culture experiments, susceptibility rankings are as follows: —, 0 to <2.0; +, 2.0 to <5.0; ++, 5.0 to <10.0; +++, ≥10.0.

^b In the cell, 3' processing is the rate-limiting step for integration (7); therefore, we either calculated the overall integration efficiency by multiplication of the enzyme efficiency (EE) values for 3' processing and strand transfer (if the enzyme efficiency for 3' processing was higher than the strand transfer efficiency) or retained the enzyme efficiency for 3' processing (if the efficiency of 3' processing was much lower than the strand transfer efficiency). ND, not determined; RC, replication capacity.

tested the susceptibilities of the various recombinant enzymes to the clinically relevant INSTIs DTG, RAL, and EVG. Molecular modeling was used to provide an understanding of interresidue and integrase-DNA interactions that drive the differential impacts of G118R, H51Y, and E138K in different HIV-1 subtypes.

Of all the recombinant proteins tested, the subtype B G118R integrase had the greatest loss of enzyme efficiency relative to that of the WT (~90% for strand transfer and ~50% for 3' processing), perhaps explaining why G118R is not selected in subtype B viruses (15). In both the CRF02_AG and subtype C proteins, the



FIG 5 Multiple-sequence alignment of subtype B, subtype C, and CRF02_AG integrase sequences from plasmids pNL4_3, pIndieC, and p97, respectively, performed by using ClustalW (v1.8). The catalytic-site residues D64, D116, and E152 are highlighted in yellow. Perfectly conserved residues are marked with asterisks, conservative substitutions are marked with : and highlighted in purple, and semiconservative substitutions are marked with • and highlighted in light blue. Nonconservative substitutions are highlighted in red. Positions E92, G140, Y143, Q148, and N155, implicated in primary resistance to RAL and EVG, are boxed in purple. Labeled residues have been adequately characterized as affecting DTG susceptibility in this study (white text with black highlighting) or in previous studies (blue boldface type) (16, 17, 37, 42).

G118R mutation alone resulted in ~70% and ~60% losses of WT strand transfer efficiency, respectively. In all subtypes tested, the G118R mutation involves a G→C transversion that is less common than the G→A transition that is seen in the case of many drug resistance mutations such as R263K (38). In subtype B, the addition of H51Y or E138K, neither of which has a major impact on strand transfer and 3' processing, partially rescued G118R-containing enzymes while not significantly increasing the activities of G118R-containing subtype C or CRF02_AG enzymes (Fig. 1B and C and 2B and C). Thus, the impact of H51Y and E138K may be related to resistance, particularly since H51Y emerged together with G118R in subtype C DTG selections.

The highly subtype-dependent protein activities revealed here by enzyme activity assays show that minor polymorphic differences among subtypes may play important roles in fitness and in pathways for resistance. A key experiment (Fig. 3) showed that G118R in subtype C caused minimal if any resistance to MK-2048 (FCs of ~2.0 and 0.87 for cell-free and cell culture assays, respectively), but the addition of E138K significantly augmented levels of such resistance (FCs of ~11 and 139, respectively). This was important since MK-2048 was the first compound to select sequentially for G118R and E138K in a subtype C viral backbone (20). More relevant clinically, the G118R mutation caused a FC in resistance to DTG of ~1.8 to 3.5 for all three subtypes (Fig. 4A) and resulted in various levels of cross-resistance to RAL (Fig. 4B) and EVG (Fig. 4C). In CRF02_AG integrase, G118R caused high-level resistance to RAL (FCs of ~8 and 17 in cell-free and cell culture assays, respectively), consistent with a clinical report of G118R in a patient harboring CRF02_AG virus who failed a RAL-containing regimen (35).

The impact of H51Y and E138K on resistance, whether alone or together with G118R, varied depending on subtype (Fig. 4). Based on integrase homology modeling and structural overlays, the G118R substitution inhibits strand transfer activity by sterically hindering the access of target DNA to the strand transfer hot spot (Fig. 6, yellow dotted circles) and by impeding hydrogen bond formation with D116. This effect was not seen with G118R in CRF02_AG (Fig. 6B) or subtype C (Fig. 6C), nor is there interference with cation coordination or interactions with D64, the most important catalytic residue (39). In contrast, 118R in subtype B forms strong electrostatic interactions with D64, C65, as well as D116 (Fig. 6A). Subtype B 118R also extends in a manner that might force the repositioning of active-site Mg^{2+} ions. In a subtype B DTG-bound model, the G118R side chain was shown physically overlap the DTG binding site (40). The presence of either H51Y or E138K caused the partial (H51Y) or complete (E138K) repositioning of the G118R side chain (Fig. 7J). In the DTG-bound models, G118R formed salt bridge interactions with D64 and hydrogen bonding with D116 in both CRF02_AG (Fig. 7H) and subtype C (Fig. 7I). In these cases, the active site would be seriously affected, and the repositioning of at least one active-site Mg^{2+} ion would be necessary (Fig. 7 H and I [potential Mg^{2+} movement is indicated by the dotted red arrow]). In both scenarios, the long side chain of 118R is completely within the DTG binding pocket (Fig. 7K and L). The addition of either H51Y or E138K in a G118R background partially (E138K) or completely (H51Y) repositioned the 118R side chain out of the DTG binding pocket in CRF02_AG (Fig. 7K). In subtype C, the addition of either H51Y or E138K was insufficient to significantly reposition the 118R side chain. This result, coupled with the slightly better activity profile caused by either substitution, may explain the

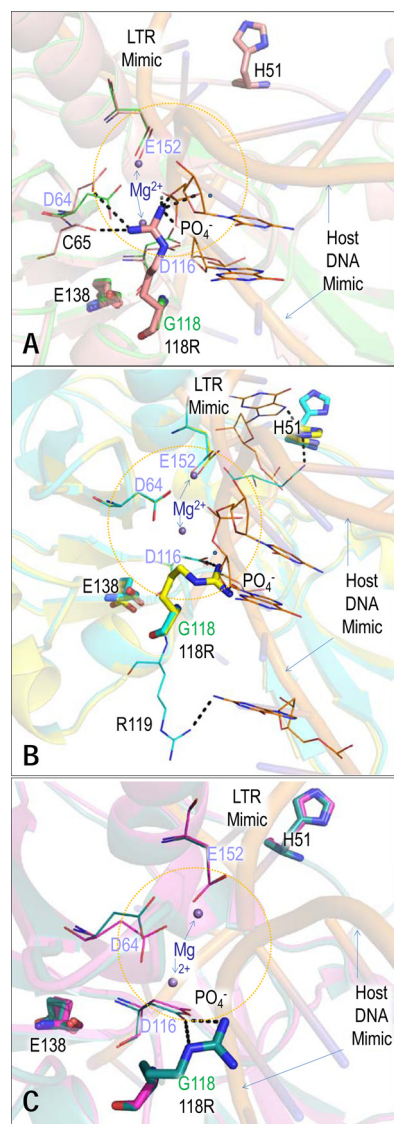


FIG 6 Modeled HIV-1 WT and G118R target capture complexes showing the differential impact of G118R on the active sites of the three integrase subtypes. HIV-1 WT or G118R monomeric homology models for each of the three subtypes were created based on the structure of the freeze-trapped PFV target capture complex (PDB accession number 4E7K). These models were then used to build dimers, as described in Materials and Methods. Integrase-DNA interaction as well as cation binding in the TCC were mimicked by direct overlay of the LTR DNA, target DNA, and the Mn^{2+} and Zn^{2+} ligands from the PFV structure. The impact of the G118R substitution on the strand transfer reaction with the active site of subtype B (A), CRF02_AG (B), and subtype C (C) was analyzed by visual assessment of the differing side-chain and backbone interactions as well as residue-DNA clashes that occur with the G118R substitution, particularly within the putative strand transfer zone (yellow circles). Structural visualization and manipulation were performed by using PyMOL. Protein and DNA structures are shown as cartoons, with the residues under investigation being shown as sticks. The catalytic triad residues are labeled in blue and shown as line traces. Structures of active-site residues interacting with residues 118, 51, and 138 are shown as line traces. Possible hydrophilic interactions between atoms separated by <3.5 Å are shown by black dashed lines. Colors of lines and sticks are based on main-chain color as well as standard atomic coloration. For clarity and where necessary, the colors of residue labels are the same as those in the cartoons for that particular model.

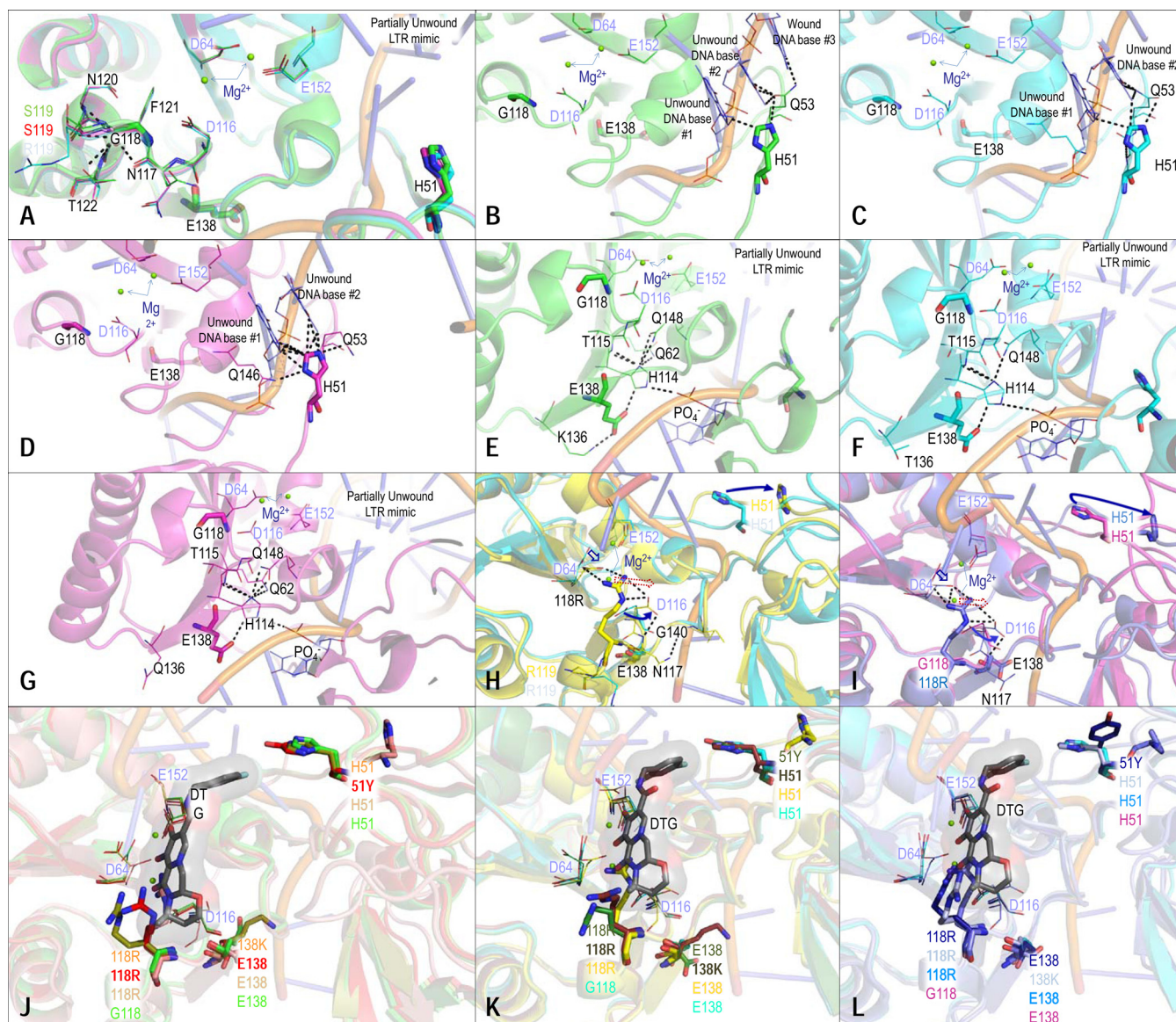


FIG 7 Active-site modeling of DTG-bound integrase. For each subtype, HIV-1 WT, G118R, H51Y/G118R, or G118R/E138K monomeric models were created based on the structure of the DTG-bound PFV structure (PDB accession number 3S3M). These models were then used to build dimers as described in Materials and Methods. Integrase-DNA interaction, cation binding, and DTG-integrase interactions were mimicked by direct overlay of the LTR DNA, the Mn^{2+} and Zn^{2+} ligands, and DTG from the PFV structure reported under PDB accession number 3S3M. (A) Overlay of WT models for subtype B (green), subtype C (pink), and CRF02_AG (turquoise) showing key backbone interactions of G118. (B) Closeup view of the subtype B WT model showing interresidue and DNA-integrase interactions of H51. (C) Closeup view of the CRF02_AG WT model showing interresidue and DNA-integrase interactions of H51. (D) Closeup view of the subtype C WT model showing interresidue and DNA-integrase interactions of H51. (E) Closeup view of the subtype B WT model showing interresidue and DNA-integrase interactions of E138. (F) Closeup view of the CRF02_AG WT model showing interresidue and DNA-integrase interactions of E138. (G) Closeup view of the subtype C WT model showing interresidue and DNA-integrase interactions of E138. (H) Overlay of WT (turquoise) and G118R (yellow) active sites showing the impact of the G118R substitution on the CRF02_AG active site. Changes in the D64 side-chain orientation caused by 118R are indicated by an open blue arrow, and other key side-chain orientation changes are indicated by a solid blue arrow; the possible repositioning of at least one Mg^{2+} cation is indicated by a dotted red arrow. (I) Overlay of WT (pink) and G118R (purple) active sites showing the impact of the G118R substitution on the subtype C active site. Changes in the D64 side-chain orientation caused by 118R are indicated by an open blue arrow, and other key side-chain orientation changes are indicated by a solid blue arrow; the possible repositioning of at least one Mg^{2+} cation is indicated by a dotted red arrow. (J) Overlay of subtype B models of WT (green), G118R (tan), H51Y/G118R (red), and G118R/E138K (orange) integrase proteins showing the positioning of the three residues H51Y, G118R, and E138K relative to DTG. (K) Overlay of CRF02_AG models of the WT (turquoise), G118R (yellow), H51Y/G118R (olive green), and G118R/E138K (tan) integrase proteins showing the positioning of the three residues H51Y, G118R, and E138K relative to DTG. (L) Overlay of subtype C models of WT (pink), G118R (purple), H51Y/G118R (navy), and G118R/E138K (light blue) integrase proteins showing the positioning of the three residues H51Y, G118R, and E138K relative to DTG. Structural visualization and manipulation were performed by using PyMOL. Protein and DNA structures are shown as cartoons, with the residues under investigation being shown as sticks. The catalytic triad residues are labeled in blue and shown as line traces. Structures of active-site residues interacting with residues 118, 51, and 138 are shown as line traces. Possible hydrophilic interactions between atoms separated by <3.5 Å are shown as black dashed lines. Colors of lines and sticks are based on main-chain color as well as standard atomic coloration. For clarity and where necessary, the colors of residue labels are the same color as those in the cartoon for that particular model.

slight increase in resistance associated with these secondary mutations.

The variable impacts of H51Y and E138K on integrase activity and drug resistance might be due to proximity with various polymorphic positions (Fig. 5). Position 51 is clearly important for DTG resistance; H51Y was a secondary resistance mutation to R263K, augmented resistance to DTG (17), and was also selected together with G118R or alone by DTG (16). The adjacent position M50 is polymorphic among subtypes, and M50I, which is a secondary substitution to R263K in subtype B (16), was shown to augment R263K-associated resistance to DTG and EVG (37). Because DTG binds primarily to DNA and does not have as many protein contacts as RAL, and because it is more flexible than EVG (40), DNA binding may be impacted by a number of the DTG-selected substitutions: H51Y (LTR DNA), G118R (target DNA), M50I (LTR DNA), and R263K and E138K (both LTR and target DNA) (16, 17, 19, 37, 41). In the case of G118R and associated substitutions, slight active-site perturbations caused by 118R may be linked. Most DTG-associated mutations appear to interact with the loop spanning L46 to T66, which spans the DNA binding trough and interacts with both C-terminal and active-site residues.

The concomitant appearance of A49P, V234L, and S119R substitutions in a previously DTG-susceptible N155H-containing subtype B virus was associated with clinical resistance to DTG (42). These substitutions propelled a significant shift in the highly conserved loop region spanning L46 to T66, drastically altering the active site and resulting in a severely compromised virus that was also highly resistant to DTG (42). M50I is the sole polymorphic residue in this stretch, and its impact on residue 49 as well as on residue 51 may vary between subtypes. The presence of multiple subtype-specific polymorphisms in association with E138K (Fig. 5) may not affect overall protein structure (Fig. 6), but altered side-chain interactions (Fig. 7) may be sufficient to cause a differential impact on DNA binding as well as the G118R-associated DTG resistance reported here. Modeling studies suggest that E138K in a DTG-bound state communicates with G118G/R primarily through the interaction of Q137 with the backbone amide of residue 118. E138 also interacts indirectly with INSTI resistance position Q148 by sequential hydrogen bonding with H114 (43). The charged nature of H114 implies that the presence of 138K may affect the H114 orientation and, ultimately, the orientation of Q148, with variable consequences for INSTI cross-resistance. The fact that E138K, in addition to having electrostatic interactions with H114, also forms such interactions with E136 in subtype B (residue 136 is T in CRF02_AG and Q in subtype C) (Fig. 5 and 7) implies that the impact of E138K on the orientation of Q148 may differ among subtypes and that the impact on drug susceptibility might also differ among subtypes.

The classical G140, Y143, and Q148 substitutions associated with resistance to RAL and EVG are in the active-site loop, and N155, also associated with resistance to RAL and EVG, is located deep in the active site. Any changes to these residues directly impact the ability of the active site to interact with the drug versus the substrate and can directly influence resistance and enzyme activity without the need for significant interresidue interactions. In these cases, the subtype may be irrelevant to the level of drug resistance that results. The EVG E92Q mutation, like G118R, is on the periphery of the INSTI binding pocket, and variable interactions involving HIV subtypes can affect the level of resistance that is associated with this substitution (44).

DTG is able to evade these primary EVG and RAL substitutions because it forms hydrophobic stacking interactions with the terminally processed viral LTR CA dinucleotide (40) and coordinates optimally with active-site Mg^{2+} ions without the need to interact significantly with the integrase protein (45). Thus, it is more difficult to develop resistance to DTG. There is growing evidence that G118R and R263K are two mutations that can engender resistance to DTG (16–19, 37, 41, 46), and both substitutions have also been shown to engender INSTI resistance in simian immunodeficiency virus (47). In this context, minor polymorphisms that vary among subtypes may be able to play a major role in the development of drug resistance.

In all three subtypes tested here, G118R results in highly deficient integration activity, and the secondary substitutions E138K and H51Y are likely primarily rescue substitutions, although their effectiveness varies between different integrase sequences. This work highlights the need to better characterize polymorphic integrase positions, particularly in treatment-experienced patients undergoing DTG therapy and especially in patients failing therapy without previously reported resistance substitutions.

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